CarVision 2.1
Car Brand Classification in Tashkent

CSAI-3210 | Midterm | Fall2026 | Report updated22October2026

1. Team, repository and contributions

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GitHub repository: NOT PROVIDED YET. This report is a review draft, not a submission-ready PDF. Insert the real repository URL here before submission; no repository has been published by the assistant.

Contribution record: the participant supplied10 phone photos and product requirements. OpenAI Codex assisted with implementation, debugging, public-data collection, local training, packaging, tests and this report. Public-photo creators retain their authorship. No student coding history, label provenance or test results have been fabricated.

Task output: a list of English (brand, colour) pairs, one per visible classifiable car. Concrete model names and top-three alternatives are optional additions. The runtime is offline and requires no API keys, internet service or payment.

Readiness: a portable Windows prototype is implemented. Remaining academic gaps include a sufficiently diverse own-photo training basis, an independent phone-camera rehearsal, the repository URL/tag and confirmation of architectural uniqueness among teams. These gaps cannot be certified by a successful EXE launch.

AI disclosure: substantially AI-assisted work. The participant must understand and explain the code, data, model decisions and limitations; no AI assistance is used during the live test.

2. Data and labels

Body/model data:779 train,161 validation,416 test photos;32 model families. Sources:528 reviewed regional Commons/Flickr crops,634 Stanford Cars crops with original model annotations,194 Chinese/EV crops. Source URLs, authors, licences and hashes are recorded in data and samples manifests. Public sources currently dominate the released weights; this does not close the assignment requirement for own Tashkent data.

Make	Body-expert training crops
BMW	35
BYD	79
Chery	38
Chevrolet	222
Daewoo	64
Geely	16
Haval	37
Honda	33
Hyundai	127
Kia	17
Lada	16
Toyota	73
Zeekr	22

Colour	Train	Validation	Test
white	399	84	82
black	404	86	84
grey	785	167	164
red	633	135	133
blue	738	154	156
green	558	120	119
brown	1585	337	339
yellow	1108	236	237

Colour counts above describe the separate VCoR colour branch, not the body dataset. Silver maps to grey; beige/gold to brown; orange to yellow. Cars in the two datasets must not be summed as unique vehicles.

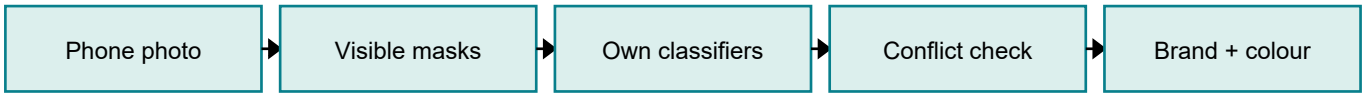
Own photos:10 originals from one outing,9 manually identified crops (Chevrolet8/Kia1; white7/black2),8 exact-model labels. Ambiguous cars remain unlabelled. All outing frames stay in TRAIN; no neighbouring view of the same car is claimed as independent validation. Their12-epoch experiment was rejected. Public source/duplicate groups retain their historical splits; complete physical-vehicle identity is unavailable for Stanford/VCoR.





Own TRAIN examples: Chevrolet Cobalt and Tracker. Same outing; not an independent test.

3. Architecture and decision rules



Stage	Operation
Phone JPEG -> laptop	Cable or messaging transfer; no external service during inference
Visible instances	General COCO Mask R-CNN; exclusive visible masks; conservative fragment union
Make and body/model	Own MobileNetV3-small avg/max-pooled heads and own references; ResNet18 body expert with aspect-preserving224padding
Colour	Own colour classifier plus visible-paint features; reflections/lighting can still cause errors
Decision consistency	Conflicting make/model outputs -> unknown make; leading model remains a question-mark hypothesis
Output	English brand colour for the assignment; JSON/TXT/annotated JPEG and optional model alternatives

ImageNet and COCO initialization are general-purpose only. No downloaded car-brand, car-model or car-colour classifier is used. ResNet18 layers3/4 and its32-class head were trained locally; the older project classifier and train-only reference features are retained as a validation-selected mixture.

Body expert parameter count: 11,192,928. Total loaded neural parameters: 63,428,707, including segmentation, make/model and colour networks. Non-neural forest/reference data are excluded. Component counts are recorded in release_v21_parameters.json.

Emblem experiment:5 manually cropped Chevrolet/Kia badges were compared using local contrast templates. Validation acceptance failed; it is disabled. Version2.1 fixes a disabled-module execution bug and a missing-brand crash. No badge-based reliability claim is made.

A high model score and a high make score come from different distributions. They are not directly comparable percentages of truth. Version2.1 discloses disagreement instead of silently displaying a conflicting confirmed make. Colour does not hard-exclude a model.

4. Training and reproducibility

Item	Recorded configuration
Main expert	Generic ImageNet ResNet18 initialization; own32-family head
Loss	Cross entropy; label smoothing0.04
Optimiser	AdamW, weight decay0.02; cosine decay
Learning rates	layer3:0.000035; layer4:0.00008; head:0.0007
Batch / hardware	24 / local CPU,3PyTorch threads
Schedule	First experiment stopped after9epochs to fix crop alignment; corrected12epochs; selected epoch6
Mixture	Expert weight 0.75; temperature 2.0
Corrected training time	29.5 minutes; measured script time
Augmentation	Mirrors, colour jitter, greyscale, blur, visible masks, upper-body crops and40-96pixel images

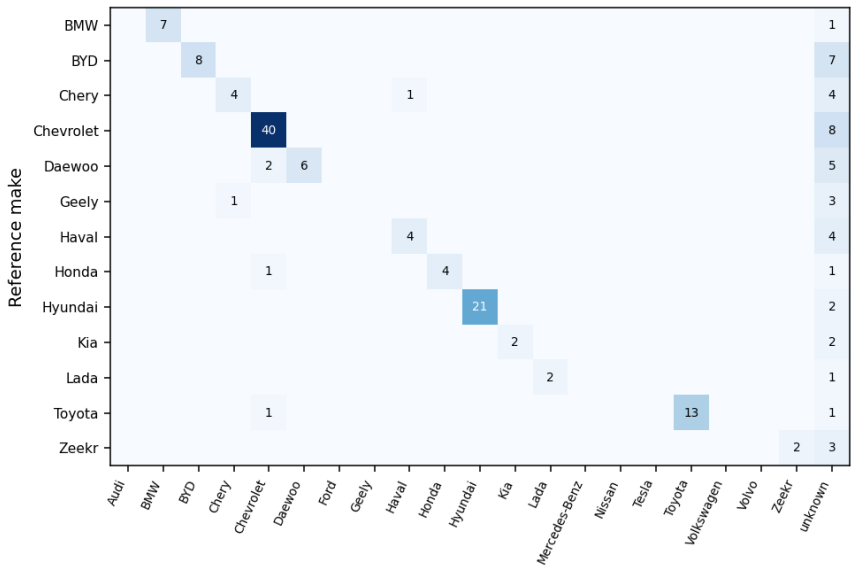
Selection used existing grouped validation only. Guards limited loss of full-crop top1/top3 and fragment top3 while improving Chevrolet macro accuracy. Calibration also uses validation. Existing test images were evaluated after selection; they have been used in prior project iterations and are not a newly blinded examination.

Own-photo experiment: frozen project-trained features and12 FC-training epochs, public rehearsal plus8 owner model crops. No candidate passed the validation guards. These rejected weights were not deployed; neither quantity nor quality of own training data is sufficient to claim the academic data requirement is complete.

Training interface: labels include image, box, brand, colour, source and group. Whole sessions/duplicates stay together. Version2.1 writes models/own_training_candidate; it no longer overwrites the production models immediately. Checkpoint, evaluation and detector training were exercised offline with a synthetic engineering fixture; those results are not recognition accuracy.

5. Evaluation and error analysis

Current make decision,161 validation body crops: 70.2% correct; 42 unknown outputs, including 20 head conflicts. Reference classes are the makes available in this body validation set, not every Tashkent brand. No detector AP or full-scene accuracy is implied.



Model top1,416 held-out crops	v1.9	v2.0/2.1 weights
whole	75.7%	81.5%
small64	34.1%	71.2%
upper55	45.4%	54.6%

Unchanged colour components: VCoR validation 94.2%, test 93.6%; masked-paint component validation 93.8%, test 92.1%. These component/crop measurements do not establish the colour accuracy of a complete scene.

Chinese models remain difficult: similar SUV silhouettes, few independent samples, reflections and unfamiliar generations cause wrong candidates. Examples include BYD/Geely/Chery confusion and white paint judged brown under warm lighting. New consistency logic removes contradictory confirmed labels; it does not retrain the model or promise100% accuracy.17 actual development/sample photos ran without exceptions. Phone-camera screen-to-laptop rehearsal has NOT been conducted.

6. How to run; 7. Reflection

Unzip the complete portable archive on Windows10/11 x64 into a writable folder. Keep CarVision.exe, _internal, models, data and samples together. Python is not required for inference. Open a phone photo, click Analyze, then copy the English brand-colour answer. Use mouse region selection only when appropriate; the exam output must cover every classifiable car.

Source setup: Python3.11; install requirements-lock.txt, then run python launch.py gui. Training: python launch.py train --labels data/labels.jsonl --output models/own_training_candidate. Evaluation uses the same labels, saved split and dataset fingerprint. Do not treat examples or development photos as new independent tests.

Engineering checks:78 automated tests including candidate-training GUI routing; end-to-end offline synthetic training/checkpoint/evaluation/detector smoke; actual sample inference regressions. Release checks also cover frozen-source correspondence, a copied EXE without Python in PATH, offline prediction, ROI, exports and archive hashes. See release logs for verification results.

Live-test procedure: photograph the displayed image with the participant phone, transfer to the laptop, run once, write the model output within2minutes. The required20 unseen photos, including at least half Chinese-brand scenes, must remain unseen. A successful local copy test is not a second-physical-computer certification.

Reflection: the weakest areas are data diversity, night/reflection colour, occluded vehicles and open-world Chinese models. Better data and independent vehicle/session validation are necessary. Increasing displayed percentages or memorising a supplied image would not solve these problems. Current own-photo experiments were honestly rejected rather than promoted.

Before submission: insert the real GitHub URL on page1; upload code, labels/data or lawful links, weights or links, environment and this corrected report; create an authentic final midterm tag; verify a clean laptop and phone rehearsal; confirm architecture uniqueness with the instructor; ensure the participant can explain the code. Deadline27October2026,23:59Tashkent; live test28October. The report is intentionally marked DRAFT while these steps remain.

References: original CSAI-3210 assignment; Stanford Cars (Krause et al.,2013); VCoR data provenance in project manifests; Wikimedia/Flickr per-file sources in THIRD_PARTY.md and samples source JSON; torchvision general ImageNet/COCO architectures. Public photographs are not credited to the participant.